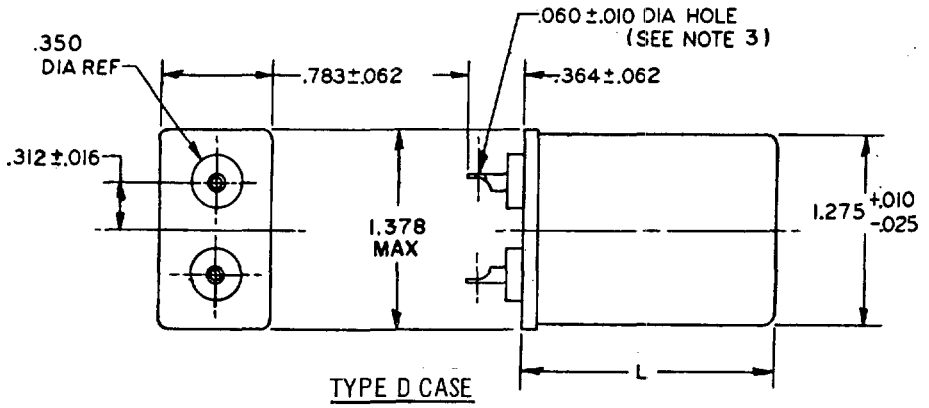
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (NONPOLARIZED, PLAIN FOIL), 85°C (VOLTAGE DERATED TO 125°C), STYLE CL52

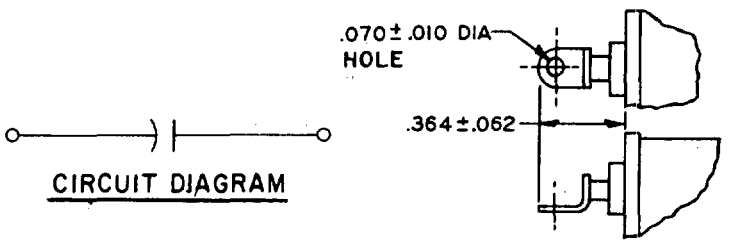
INACTIVE FOR NEW EQUIPMENT DESIGN. NO SUPERSEDING DOCUMENT.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.



Dimensions	
	L ±.062 (1.57)
D1	1.375 (34.93)
D2	2.000 (50.80)
D3	2.500 (63.50)



CIRCUIT DIAGRAM

OPTIONAL ALTERNATE TERMINAL CONSTRUCTION

INCHES	MM	INCHES	MM
.010	.25	.312	7.92
.016	.41	.350	8.89
.025	.64	.364	9.25
.060	1.52	.783	19.89
.062	1.57	1.275	32.39
.070	1.78	1.378	35.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Orientation of terminal hole not fixed.

Ⓗ FIGURE 1. Style CL52 capacitors.

Ⓗ denotes changes

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Ⓗ TABLE I. Capacitor characteristics.

Type designation	Voltage			Capacitance	DC leakage (max)			Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C	125°C		
	volts, dc	volts, dc	volts, dc	μF	μA	μA	μA	ohms	
CL52BE421MNG	15	10	17.2	420	42	335	420	6.6	D2
CL52BJ650MNG	50	30	57.5	65	19	150	190	42.7	D1
CL52BQ400MNG	150	100	172.0	40	40	320	400	69.5	D2

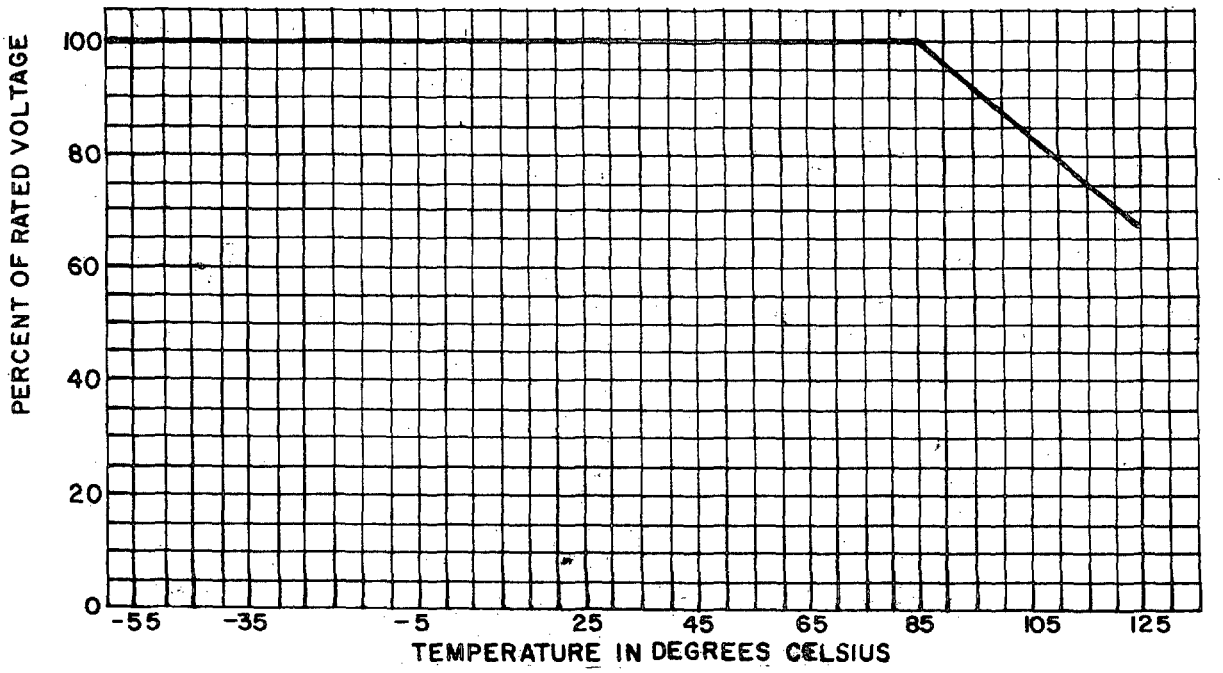


FIGURE 2. Voltage derating with temperature.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration -
Capacitors - See figure 1.
Brackets - See figure 3.
Case type - Rectangular, uninsulated.
Seal type - Hermetic (symbol E).
Terminals - See figure 1.
Rated temperature - -55° C to +85° C, voltage derated to +125° C (see figure 2).
Rated voltage - See table I.
Capacitance (cap) (nom) - See table I.
Capacitance tolerance -
Symbol M (±20%).

DC leakage (DCL) (max) (at +25° C):
See table I.

Dissipation factor (DF) (max): 10%

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - 2, 000 vdc.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min): 100 megohms.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - 115% of initial requirement.

Solderability: Not applicable.

Terminal strength:
Pull test - In accordance with MIL-C-3965.
Bend test (if req.) - In accordance with MIL-C-3965.
Torque test - Not applicable.
Wire-lead bend test - Not applicable.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±3% of initial measurement.
DF (max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See table I (25° C).
ΔCap - Within ±3% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25° C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DF (max) - Shall not exceed initial requirement.
Step 2 (-55° C):
Impedance - See table I.
ΔCap - Within -35% of step 1 value.
Step 3 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±3% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85° C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - 25%.
Step 5 (+125° C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - 25%.
Step 6 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±3% of initial measurement.
DF (max) - Shall not exceed initial requirement.

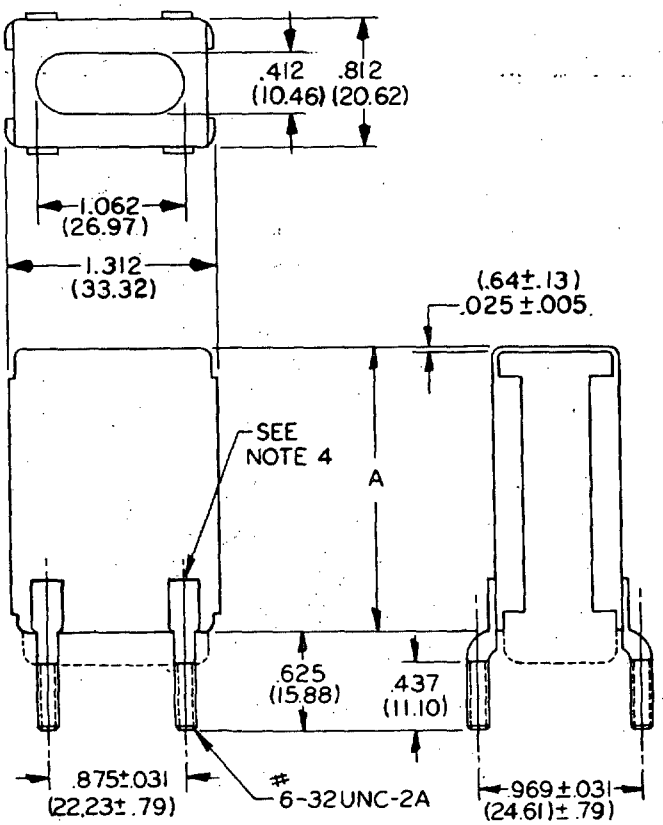
Reverse voltage: Not applicable.

Life at +85° C:
2, 000-hr (qualification):
DCL (max) - See table I (25° C).
ΔCap - Within +15, -20% of initial measurement.
DF (max) - 130% of initial requirement.
240-hr (group B):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - 130% of initial requirement.
1, 700-hr (group C):
DCL (max) - See table I (25° C).
ΔCap - Within +15, -20% of initial measurement.
DF (max) - 130% of initial requirement.

Barometric pressure: Rated voltage.

Life at +125° C:
DCL (max) - See table I (25° C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 150% of initial requirement.

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BRACKETS FOR TYPE D CASE

Type designation	Case size	Dimensions						
		Dim	Inches			Millimeters	
			Min	Nom	Max	Min	Max
CL050SD13	D1	A	1.159	1.175	1.191	29.44	30.25
CL050SD23	D2	A	1.784	1.800	1.816	45.31	46.13
CL050SD33	D3	A	2.284	2.300	2.316	58.01	58.82

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Spade lugs shall be rigid (either double welded, double riveted, single riveted and soldered, or single riveted and welded to the bracket).
5. Unless otherwise specified, tolerance is ±.062 (1.57 mm).

FIGURE 3. CL050 bracket.

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Marking: Capacitors shall be marked as shown in the following example:
Example:
CL52BE211MNG - Type designation.
12345 7015 - Source and date code.
210 μF 15 vdc - Capacitance and rated voltage (at 85° C).

Voltage groups (85° C rated voltage):
I - - - - - - - - - - - - 15 to 50 v, incl.
II - - - - - - - - - - - - 75 to 150 v, incl.

Substitution data. Capacitors covered by this specification are directly substitutable for capacitors of equivalent capacitance, capacitance tolerance, and voltage in the same case size.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Navy - SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Air Force - 85

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